

Eichler–Shimura defects of meromorphic period polynomials on the winding-class lattice

A computational session note

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Abstract

This note documents one session of work on the canonical-contour problem (item T1 of the pre-arXiv agenda): which winding classes of the two-segment contour give period polynomials of meromorphic modular forms satisfying the Eichler–Shimura relations $r|(1+S) = 0$ and $r|(1+U+U^2) = 0$. Everything below concerns the period polynomial of the *actual meromorphic form*, computed by direct numerical integration of f along the two-segment contour; no pole subtraction or projection enters any computation. The main outcomes: (i) two exact identities of the Hurwitz T -kernel — a reflection identity forcing $\mathbf{K}_T|(1+S) = 0$, and the fact that at integer s the kernel is a Bernoulli polynomial — which together reduce the whole problem to integer linear algebra with closed-form entries; (ii) the “the $(1+S)$ -defect of an i -pole is nonzero in every class” conjecture is *false*: an explicit class kills it (verified to 10^{-33}); (iii) for the weight-4 elliptic block, the classes satisfying *both* relations are completely characterized (Proposition 3): a nonempty rank-4 lattice coset, every weight in the defining equations derived by hand; (iv) in the clean class the original resonance proposition is restored and now hand-derived: the pole breaks exactly its resonant relation, by exactly its residue. Open items and all caveats are listed in §7.

1 Setting

Conventions follow the companion note [1]; this section makes the present note self-contained. Fix even weight k and $n = k - 2$, and let $\Gamma = \mathrm{PSL}_2(\mathbb{Z}) = \langle S, T \rangle$ act on $\mathbb{H}$ by $S\tau = -1/\tau$, $T\tau = \tau + 1$. A *meromorphic modular form* of weight k is a quotient $f = f_1/f_2$ with $f_i \in M_{k_i}$ holomorphic modular, $f_2 \not\equiv 0$, $k = k_1 - k_2$; we write F_k for the space of these. Let $\Gamma \cdot f^{-1}(\infty)$ denote the Γ -orbit of the poles of f in $\mathbb{H}$. For $\tau_0 \in \mathbb{H}$, the *S- and T-segments* are smooth paths in $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$,

$$\gamma_{\tau_0}^S : -1/\tau_0 \longrightarrow \tau_0, \quad \gamma_{\tau_0}^T : \tau_0 - 1 \longrightarrow \tau_0,$$

and the *L-integral* of $f \in F_k$ along the pair is

$$L^*(f, s) = e^{-\pi i s/2} \left[\int_{\gamma_{\tau_0}^S} f(\tau) \tau^{s-1} d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \tilde{k}(\tau, s) d\tau \right], \quad \tilde{k}(\tau, s) = \zeta(1-s, \tau+1) e^{i\pi(s-1)} \zeta(1-(k-s), \tau+1), \quad (1)$$

with $\zeta(s, x)$ the Hurwitz zeta function. Both integrals are finite for every $s \in \mathbb{C}$ (compact contours avoiding the poles).

Proposition 1 (Class dependence only [1]). $L^*(f, s)$ depends on $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ only through the homotopy class of the pair in $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$; within a class it is independent of τ_0 and of the paths. For f with no interior poles there is a single class and L^* is canonical.

Proof sketch. Path independence within a class is Cauchy's theorem (the integrands are holomorphic off the poles). Independence of τ_0 within a chamber follows because the boundary terms of $\partial_{\tau_0} L^*$ cancel between the two segments, using only T -periodicity $f(\tau_0 - 1) = f(\tau_0)$ and modularity $f(-1/\tau_0) = \tau_0^k f(\tau_0)$; see [1]. $\square$

Proposition 2 (Wall-crossing). Record a class by its winding data (X_S, X_T) : integers, one pair per pole, counting the net turns of γ^S , resp. γ^T , about each pole relative to a reference class with value L_0^* . Then

$$L^*(f, s; X_S, X_T) = L_0^*(f, s) + 2\pi i e^{-\pi i s/2} \sum_p \left[X_S(p) \operatorname{Res}_{\tau=p}(f \tau^{s-1}) + X_T(p) \operatorname{Res}_{\tau=p}(f \tilde{k}(\tau, s)) \right]. \quad (2)$$

Proof. Two pairs whose winding data differ by one unit at a single pole p on one segment bound a loop encircling p once; the residue theorem adds $2\pi i$ times the residue of that segment's integrand, and the prefactor $e^{-\pi i s/2}$ multiplies through. Sum over crossings. $\square$

The period polynomial (overall $(2\pi i)^{n+1}$ dropped, as in the companion scripts) is

$$r_f(X, Y) = \sum_{\ell=0}^n c_\ell X^{n-\ell} Y^\ell, \quad c_\ell = (-1)^\ell \binom{n}{\ell} i^{\ell+1} L^*(f, \ell+1), \quad (3)$$

equivalently $r_f = \int_{\gamma_S} f(X - \tau Y)^n d\tau + \int_{\gamma_T} f \mathbf{K}_T d\tau$ with the assembled kernel

$$\mathbf{K}_T(\tau; X, Y) = \sum_{\ell=0}^n \binom{n}{\ell} (-1)^\ell \tilde{k}(\tau, \ell+1) X^{n-\ell} Y^\ell. \quad (4)$$

Slash actions: $(P|S)(X, Y) = P(-Y, X)$, $(P|U)(X, Y) = P(X - Y, X)$, $(P|U^2)(X, Y) = P(-Y, X - Y)$. Define the two *defect operators*

$$D_S(P) := P|(1 + S), \quad D_U(P) := P|(1 + U + U^2).$$

For cusp forms and weakly holomorphic forms, $D_S(r_f) = D_U(r_f) = 0$ (proven in [1]; verified there against Brown's quasi-periods [2] and, via Haberland's formula [3], against Petersson norms). The question is what replaces this for meromorphic f .

2 The winding lattice and the linear defect model

By Propositions 1 and 2, the value of L^* depends on the class only through the winding data (X_S, X_T) . At the integer values $s = \ell + 1$ entering (3), one crossing of pole p by γ^S shifts r_f by $2\pi i R_p$, and one crossing by γ^T shifts it by $2\pi i Q_p$, where

$$R_p := \operatorname{Res}_{\tau=p}(f(\tau)(X - \tau Y)^n), \quad Q_p := \operatorname{Res}_{\tau=p}(f(\tau) \mathbf{K}_T(\tau; X, Y)). \quad (5)$$

Hence, writing $x = (x_p)$, $t = (t_p)$ for the winding integers of a class relative to a base class with polynomial $r_f^{(0)}$,

$$r_f(x, t) = r_f^{(0)} + 2\pi i \sum_p x_p R_p + 2\pi i \sum_p t_p Q_p, \quad (6)$$

and the defects D_S, D_U are *affine-linear in the winding integers*. Every question of the form “is there a class with $D_S(r_f) = 0$ and/or $D_U(r_f) = 0$ ” is therefore an integer linear-algebra question with coefficient vectors $D_S(R_p), D_U(R_p), D_S(Q_p), D_U(Q_p)$. This specializes Proposition 2 to integer s ; the specialization was verified numerically to 32 digits.

3 Two exact identities of the T -kernel

Lemma 1 (Reflection). *For even k and integers $0 \leq m \leq n$,*

$$\tilde{k}(\tau, k - 1 - m) = -(-1)^m \tilde{k}(\tau, m + 1).$$

Proof. At $s = m + 1$ the phase is $e^{i\pi m} = (-1)^m$, so $\tilde{k}(\tau, m + 1) = \zeta(-m, \tau + 1) - (-1)^m \zeta(m + 2 - k, \tau + 1)$. At $s = k - 1 - m$ the two Hurwitz arguments are $1 - s = m + 2 - k$ and $1 - (k - s) = -m$, and the phase is $e^{i\pi(k-2-m)} = (-1)^m$ for even k ; so $\tilde{k}(\tau, k - 1 - m) = \zeta(m + 2 - k, \tau + 1) - (-1)^m \zeta(-m, \tau + 1)$, which is $-(-1)^m$ times the previous display. $\square$

Corollary 1. $\mathbf{K}_T|S = -\mathbf{K}_T$, hence $D_S(\mathbf{K}_T) = 0$ identically, and therefore:

(i) $D_S(Q_p) = 0$ for every pole p and every f : T -windings never move the $|(1 + S)$ -defect.

(ii) At $\tau_0 = i$ the S -segment degenerates to a point, so $r_f = \int_{\gamma_T} f \mathbf{K}_T$ and $D_S(r_f) = \int_{\gamma_T} f D_S(\mathbf{K}_T) = 0$: the S -relation holds automatically, for every $f \in F_k$ with no pole on the contour, in the $\tau_0 = i$ class.

Proof. In (4) substitute $(X, Y) \rightarrow (-Y, X)$ and re-index $\ell \rightarrow n - \ell$; the coefficient of $X^{n-\ell} Y^\ell$ in $\mathbf{K}_T|S$ is $\binom{n}{\ell} \tilde{k}(\tau, k - 1 - \ell)$, which by Lemma 1 equals $-(-1)^\ell \binom{n}{\ell} \tilde{k}(\tau, \ell + 1)$, i.e. minus the coefficient in $\mathbf{K}_T$. Items (i), (ii) follow because D_S commutes with $\text{Res}_p(f \cdot)$ and with $\int_{\gamma_T} f \cdot$. $\square$

Corollary 1(ii) upgrades an empirical observation (all five test forms held $|(1 + S)$ to $\sim 10^{-33}$ at $\tau_0 = i$, evidence item [E4] of the working ledger) to a theorem, and (i) decouples the problem: the S -defect is controlled by γ^S -windings alone, while γ^T -windings move only the U -defect.

Lemma 2 (Bernoulli form of the kernel). *At the integer points $s = m + 1$, $0 \leq m \leq n$, using $\zeta(-j, x) = -B_{j+1}(x)/(j + 1)$,*

$$\tilde{k}(\tau, m + 1) = -\frac{B_{m+1}(\tau + 1)}{m + 1} + (-1)^m \frac{B_{k-1-m}(\tau + 1)}{k - 1 - m},$$

a polynomial in τ . In particular, for $k = 4$:

$$\tilde{k}(\tau, 1) = -B_1(\tau + 1) + \frac{1}{3}B_3(\tau + 1), \quad \tilde{k}(\tau, 2) = -B_2(\tau + 1), \quad \tilde{k}(\tau, 3) = -\tilde{k}(\tau, 1),$$

$$\mathbf{K}_T = \tilde{k}(\tau, 1) (X^2 - Y^2) + 2 B_2(\tau + 1) XY. \quad (7)$$

Consequently *all* shift vectors (5) are residues of f against explicit polynomials: the canonical-class problem has closed-form integer data. (The visible $(X^2 - Y^2)$ and XY in (7) are each killed by $(1 + S)$ — Corollary 1 made concrete.)

4 The weight-4 elliptic block: hand derivations and verdicts

Let $f = E_4\Delta/E_6^2 \in F_4$, the canonical block with a double pole at i (and no other poles mod Γ), Laurent data $a_{-2} = -1/(1728\pi^2)$ and the parity relation of the following lemma.

Lemma 3 (Parity at the S -fixed point). *Let $f \in F_4$ have a double pole at i , $f = a_{-2}(\tau - i)^{-2} + a_{-1}(\tau - i)^{-1} + O(1)$. Then $a_{-1} = ia_{-2}$.*

Proof. Write $\tau = i + \varepsilon$; then $-1/\tau = i - \varepsilon - i\varepsilon^2 + O(\varepsilon^3)$, so $f(-1/\tau) = a_{-2}\varepsilon^{-2}(1 - 2i\varepsilon) - a_{-1}\varepsilon^{-1} + O(1)$, while $\tau^4 f(\tau) = (1 - 4i\varepsilon)(a_{-2}\varepsilon^{-2} + a_{-1}\varepsilon^{-1}) + O(1)$. Modularity $f(-1/\tau) = \tau^4 f(\tau)$ at order ε^{-1} gives $-2ia_{-2} - a_{-1} = a_{-1} - 4ia_{-2}$, i.e. $a_{-1} = ia_{-2}$. $\square$

4.1 Residue polynomials at the i -orbit

For a double pole, $\text{Res}_i(fg) = a_{-1}g(i) + a_{-2}g'(i)$ for holomorphic g . With $g = (X - \tau Y)^2$:

$$R_i = a_{-1}(X - iY)^2 - 2a_{-2}Y(X - iY) \stackrel{a_{-1}=ia_{-2}}{=} ia_{-2}(X^2 + Y^2). \quad (8)$$

By T -periodicity of f , $R_{i\pm 1}$ is (8) with $X \rightarrow X \mp Y$:

$$R_{i+1} = ia_{-2}((X - Y)^2 + Y^2), \quad R_{i-1} = ia_{-2}((X + Y)^2 + Y^2). \quad (9)$$

4.2 Defects of the shift vectors, by hand

Direct substitution gives

$$\begin{aligned} (X^2 + Y^2) | (1 + S) &= 2(X^2 + Y^2), & ((X \mp Y)^2 + Y^2) | (1 + S) &= 3(X^2 + Y^2), \\ (X^2 + Y^2) | (1 + U + U^2) &= 4(X^2 - XY + Y^2), & ((X - Y)^2 + Y^2) | (1 + U + U^2) &= 4(X^2 - XY + Y^2), \\ & & ((X + Y)^2 + Y^2) | (1 + U + U^2) &= 8(X^2 - XY + Y^2). \end{aligned}$$

Writing $\sigma := 2\pi i R_i = -2\pi a_{-2}(X^2 + Y^2)$ for the unit on the S -side and normalizing the U -side by $D_U(2\pi i R_i)$, the shift table is:

generator	x_i	x_{i-1}	x_{i+1}	t_i	t_{i-1}	t_{i+1}
S -defect weight (units of σ)	2	3	3	0	0	0
U -defect weight (units of $D_U(2\pi i R_i)$)	1	2	1	7/12	7/12	19/12

The zeros in the T -columns of the S -row are Corollary 1(i). The integer entries 2, 3, 3 and 1, 2, 1 are the one-line computations above. The T -column entries are derived in §4.3; every entry of the table is therefore exact. (Each was independently confirmed by PSLQ at 34 digits, off-ray residuals $\lesssim 10^{-35}$.)

4.3 The T -columns, by hand

From (7), $\text{Res}_i(f\mathbf{K}_T) = a_{-1}\mathbf{K}_T(i) + a_{-2}\partial_\tau\mathbf{K}_T(i)$, with $\tilde{k}(\tau, 1) = \frac{1}{3}\tau^3 + \frac{1}{2}\tau^2 - \frac{5}{6}\tau - \frac{1}{2}$ and $B_2(\tau + 1) = \tau^2 + \tau + \frac{1}{6}$, so that $\tilde{k}(i, 1) = -1 - \frac{7}{6}i$, $\tilde{k}'(i, 1) = -\frac{11}{6} + i$, $B_2(i + 1) = -\frac{5}{6} + i$. Using $a_{-1} = ia_{-2}$,

$$Q_i = a_{-2} \left[(i\tilde{k}(i, 1) + \tilde{k}'(i, 1))(X^2 - Y^2) + 2(iB_2(i + 1) + (2i + 1))XY \right] = a_{-2} \left[-\frac{2}{3}(X^2 - Y^2) + \frac{7}{3}iXY \right].$$

Since $(X^2 - Y^2) | (1 + U + U^2) = 0$ while $XY | (1 + U + U^2) = X^2 - XY + Y^2$,

$$D_U(2\pi i Q_i) = -\frac{14}{3} \pi a_{-2} (X^2 - XY + Y^2) = \frac{7}{12} D_U(2\pi i R_i).$$

For the translates, the Bernoulli difference equations give $\mathbf{K}_T(\tau + 1) = \mathbf{K}_T(\tau) + (\tau^2 + 2\tau)(X^2 - Y^2) + 4(\tau + 1)XY$, whence (using 1-periodicity of f)

$$Q_{i+1} - Q_i = \text{Res}_i \left(f \cdot [(\tau^2 + 2\tau)(X^2 - Y^2) + 4(\tau + 1)XY] \right) = a_{-2} [i(X^2 - Y^2) + 4iXY],$$

$$Q_{i-1} - Q_i = -\text{Res}_i \left(f \cdot [(\tau^2 - 1)(X^2 - Y^2) + 4\tau XY] \right) = 0,$$

the second vanishing by the same parity cancellation ($a_{-1}(-2) + a_{-2}(2i) = 0$ and $a_{-1}(4i) + 4a_{-2} = 0$). Hence

$$\mu_U(t_{i-1}) = \frac{7}{12}, \quad \mu_U(t_{i+1}) = \frac{7}{12} + \frac{D_U(2\pi i(Q_{i+1} - Q_i))}{D_U(2\pi i R_i)} = \frac{7}{12} + 1 = \frac{19}{12},$$

completing the table. The equality $Q_{i-1} = Q_i$ — the repeated $7/12$ — is thus a parity identity, not a numerical coincidence.

4.4 The clean class: the resonance proposition, restored

In the clean class ($\tau_0 = 0.3 + 1.4i$, straight segments, no pole enclosed), the measured defects of the actual r_f are

$$r_f | (1 + U + U^2) = 0 \text{ (residual } 5.6 \times 10^{-48}), \quad r_f | (1 + S) = -2\pi a_{-2} (X^2 + Y^2) = 2\pi i R_i, \quad (10)$$

the right-hand side now *derived* in (8) and equal to the measurement to 48 digits. Together with the mirror behaviour of ρ -pole forms (which in their clean classes hold $| (1 + S) |$ and break $| (1 + U + U^2) |$; see §5), this restores the original proposition in class-anchored form: *in the clean class, a pole on the fixed locus of an elliptic element breaks exactly its resonant relation, by exactly its residue polynomial*. What is genuinely class-dependent — and what the earlier “in every class” version missed — is quantified next.

4.5 The obstruction conjecture is false; joint classes exist

The earlier conjecture (ledger item [E5]) asserted the S -defect of an i -pole is nonzero in *every* class, based on sweeping only x_i (weight 2: even shifts of an odd base). The table refutes it: $\gcd(2, 3) = 1$, and indeed with base defect σ ,

$$x = (1, -1, 0) : \quad \sigma(1 + 2 - 3) = 0,$$

verified by direct high-precision recomputation of r_f in that class: $| (1 + S) | / \text{scale} = 7 \times 10^{-33}$ (the U -relation then breaks, as the table predicts). More is true: *both* relations can be killed simultaneously, and the solution set can be characterized completely.

Proposition 3 (Joint Eichler–Shimura exactness for the weight-4 block). *Let $f = E_4 \Delta / E_6^2$, and take as reference the clean class, whose base defect pair is $(2\pi i R_i, 0)$ (10). Among the depth-one*

winding classes $(x_i, x_{i-1}, x_{i+1}; t_i, t_{i-1}, t_{i+1}) \in \mathbb{Z}^6$, those whose period polynomial satisfies both $r_f|(1+S)=0$ and $r_f|(1+U+U^2)=0$ are exactly the integer solutions of

$$1 + 2x_i + 3x_{i-1} + 3x_{i+1} = 0, \quad 12x_i + 24x_{i-1} + 12x_{i+1} + 7t_i + 7t_{i-1} + 19t_{i+1} = 0. \quad (11)$$

This set is exactly the coset

$$(x; t) = (1, -1, 0; -1, 0, 1) + \mathbb{Z} \left\langle (0, 0, 0; 1, -1, 0), (0, 0, 0; -19, 0, 7), (-3, 2, 0; 1, 0, -1), (0, -1, 1; -1, 0, 1) \right\rangle, \quad (12)$$

a coset of a rank-4 sublattice of $\mathbb{Z}^6$. In particular Eichler–Shimura exactness holds in infinitely many classes, and exactness alone does not select a contour.

Proof. By the linear model (6) (Proposition 2 specialized to integer s), the two defects of the class $(x; t)$ are affine-linear in the winding integers with shift vectors D_S, D_U of $2\pi i R_p, 2\pi i Q_p$ at $p \in \{i, i \pm 1\}$. All twelve weights are derived above: the S -row is $(2, 3, 3; 0, 0, 0)$ in units of $\sigma = 2\pi i R_i$ (§4.2 for the integer entries via (8)–(9); the T -entries vanish by Corollary 1(i)), and the U -row is $(1, 2, 1; \frac{7}{12}, \frac{7}{12}, \frac{19}{12})$ in units of $D_U(2\pi i R_i)$ (§4.2 and §4.3). Both defect components lie on the fixed rays $\mathbb{C}(X^2+Y^2), \mathbb{C}(X^2-XY+Y^2)$, so vanishing of the defect pair is equivalent to the two scalar equations: base 1 plus the S -row for the first; base 0 plus the U -row, cleared of denominators ($\times 12$), for the second — i.e. (11). Diophantine solution. The integer solutions of the homogeneous equation $2a + 3b + 3c = 0$ are exactly $\alpha(3, -2, 0) + \gamma(0, -1, 1)$: reducing mod 3 forces $3 \mid 2a$, hence $3 \mid a$, and the parametrization follows. A particular solution of the first equation of (11) is $x^{(0)} = (1, -1, 0)$. Writing $N := x_i + 2x_{i-1} + x_{i+1}$, the second equation reads $7t_i + 7t_{i-1} + 19t_{i+1} = -12N$, with particular solution $t = N(1, 0, -1)$ (because $7 - 19 = -12$) and homogeneous solutions $s(1, -1, 0) + u(-19, 0, 7)$ (reducing mod 7 forces $7 \mid 19t_{i+1}$, hence $7 \mid t_{i+1}$). Since $N(x^{(0)}) = -1$, $N(-3, 2, 0) = 1$, and $N(0, -1, 1) = -1$, assembling particular-plus-kernel in x with the matching $t = N(1, 0, -1)$ gives exactly the coset (12); the four displayed kernel vectors generate the *full* kernel lattice, since any homogeneous solution, after subtracting multiples of the third and fourth, has x -part zero and is then a combination of the first two. Each generator checks against (11) directly, e.g. $(-3, 2, 0; 1, 0, -1)$: $-6 + 6 = 0$ and $-36 + 48 + 7 - 19 = 0$; the particular solution: $1 + 2 - 3 = 0$ and $12 - 24 - 7 + 19 = 0$. $\square$

(A float box search found 66 of these solutions with all windings in $[-3, 3]$, each consistent with (11) — the numerical shadow of the proposition. The inputs a reader may wish to interrogate: the wall-crossing law, Proposition 2, verified to 32 digits; the parity Lemma 3; and the clean-class base pair, measured to 48 digits, (10).)

Remark 1 (Why six winding integers, and not two). One might expect just two encircling numbers — one per segment — for a form with a single pole orbit. But the puncture set of $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$ is the *whole orbit*, and windings about distinct orbit points are inequivalent because the integrand weights are not translation-invariant: by (9), $R_{i \pm 1} \neq R_i$ — the same pole of f , but a different residue polynomial, which is where the distinct weights (2 vs. 3; $7/12$ vs. $19/12$) come from. The depth-one window keeps the three orbit representatives adjacent to the contour, times two segments: $\mathbb{Z}^6$. The two-winding intuition is, in fact, precisely the truncation behind the refuted conjecture of §4.4: with windings at i alone, the S -equation of (11) collapses to $1 + 2x_i = 0$, which has no integer solution — the old “parity obstruction” — and it is the translate windings, with their weight 3, that dissolve it.

5 The ρ -blocks and a generic pole

The same machinery was run on E_6/j ($k = 6$, triple pole at $\rho = e^{2\pi i/3}$), $E_8/j = \Delta/E_4$ ($k = 8$, simple pole at ρ), and the generic-pole form $E_6/(j - j(1.1i))$ ($k = 6$). Base class $\tau_0 = i$ throughout (which holds $|(1 + S)|$ identically by Corollary 1(ii)); windings allowed at the poles adjacent to the contour (“depth one”: the poles and their T -translates and S -images next to the fundamental domain), boxes $|x| \leq 3$.

E_6/j . All U -defects lie on one real ray with PSLQ-exact weights

$$\mu_U : \quad x_\rho : -11, \quad x_{\rho+1} : -3, \quad t_\rho : -\frac{8}{5}, \quad t_{\rho+1} : -\frac{48}{5}, \quad \text{base} = 1.$$

Killing U alone is feasible: $3 \cdot (-3) + 1 \cdot (-\frac{8}{5}) + (-1) \cdot (-\frac{48}{5}) = -1$, i.e. $(x, t) = ((0, 3), (1, -1))$, verified to 4.1×10^{-31} (the S -relation then breaks, weight 6.0). Killing *both* failed in the depth-one window: the two S -defect vectors $D_S(R_\rho), D_S(R_{\rho+1})$ are $\mathbb{Z}$ -independent (no rational ray), pinning $x_\rho = x_{\rho+1} = 0$; the T -windings alone then need $\frac{8}{5}t_1 + \frac{48}{5}t_2 = 1$, i.e. $8t_1 + 48t_2 = 5$ — impossible mod 8. *Caveat: this is an obstruction of the truncated window, not (yet) of the full winding lattice; deeper windings refine the reachable lattice and may dissolve it.* No impossibility proposition is claimed here or anywhere in this note: the mod-8 argument is exact only within the depth-one window and rests on the PSLQ-identified weights (not yet hand-derived, unlike the $k = 4$ table). The honest general statement is:

Question. For which blocks does the winding lattice reach joint exactness — equivalently, for which pole orbits does the defect coset contain the origin? The weight-4 i -block does (Proposition 3); the depth-one window of E_6/j does not; the full lattice for E_6/j , Δ/E_4 , and the generic pole is undecided.

5.1 A second generic pole, at depth two: $z = 0.47 + 0.99i$

To probe the generic case beyond depth one, take $f = E_6/(j - j(z))$ with $z = 0.47 + 0.99i$ ($|z| \approx 1.096 > 1$, off the imaginary axis, off the edge $\text{Im } \tau = 1$: a fundamental-domain point that avoids every special locus). Because the base class $\tau_0 = i$ holds the S -relation identically (Corollary 1(ii)), one may search using only moves that provably keep the S -defect at zero: T -windings anywhere (Corollary 1(i)), and the pairs “wind q once, wind Sq once backwards”, which are exactly S -silent since $R_{Sq} = R_q|S$ for even n (change of variables $\tau \rightarrow -1/\tau$ in (5), using $f(-1/\tau) = \tau^k f(\tau)$ and $d(-1/\tau) = \tau^{-2} d\tau$). The joint problem then reduces to a *single* complex linear Diophantine equation on the U -ray ($\dim_{\mathbb{C}} \ker(1 - U) = 1$ at $n = 4$): $\sum_g x_g \mu_g = -1$ with $\mu_g := D_U(\text{move}_g)/D_U(\text{base})$.

The computation (13 moves: 5 pairs, 8 T -windings, spanning orbit depth two; all checks passed — base S -defect 8×10^{-45} , pair S -defects $\lesssim 10^{-40}$, off-ray residuals $\lesssim 10^{-41}$) shows what genericity does and does not destroy:

- The weights μ_g are now genuinely complex and (empirically) irrational — e.g. $\mu_{\text{pair}(z, Sz)} = -26.9827 \dots + 70.1532 \dots i$ — as expected once both the isotropy and the singular-moduli arithmetic of Remark 2 are absent.
- The exact $\mathbb{Z}[\Gamma]$ -module relations survive and are confirmed to working precision (40 digits): PSLQ, asked for integer relations, first returns *structural* identities, e.g. $\mu_{\text{pair}(z, Sz)} = \mu_{T@Sz} - \mu_{T@(Sz-1)}$ and $\mu_{T@z} + \mu_{T@Sz} = \mu_{T@(z-1)} + \mu_{T@(Sz-1)}$ — Hurwitz difference-equation identities among transcendental numbers, holding on the nose. The module structure is real; only its *arithmetic normalization* was special at CM points.

- No joint-kill class was found: the search (integer box on the 13 distinct weights; then PSLQ with a greedy deflation of structural relations, maxcoeff 10^6) terminated in the relation $8 + 3\mu_{T@Sz} + 5\mu_{T@(z-1)} + 2\mu_{T@S(z-1)} + 5\mu_{T@S(z+1)} = 0$: within this window the reachable integer multiples of the base U -defect include **8**, but no combination producing 1 was found.

The recurrence of 8 is striking: the depth-one obstruction of E_6/j (§5, mod-8) and this depth-two generic-pole computation — two entirely different pole locations at the *same weight* $k = 6$ — both stall at index 8, while the weight-4 block was jointly killable through its twelfths. This raises the possibility that joint reachability is governed by the *weight* (through the Bernoulli arithmetic of the kernel) rather than, or in addition to, the isotropy of the pole. Caveats: the box was narrow, the deflation path greedy, and a single PSLQ relation certifies reachability of 8 but not unreachability of 1; “index 8” is evidence, not a theorem.

Δ/E_4 ($k = 8$). The U -defect space is genuinely multi-dimensional (no single ray; $\dim \ker(1 - U) = 3$ on degree-6 forms), rank 7 of 10 generators; no joint solution in the $|x| \leq 2$ window. Inconclusive.

$E_6/(j - j(1.1i))$ (**generic pole**). Base class: S -defect = 0 (Corollary 1(ii)), U -defect $\neq 0$. Each relation is separately killable (this reproduces the earlier routed-contour observations); rank 7 of 12 generators, real solutions of the joint system exist, but no *integer* joint solution appeared with the five free windings in $[-3, 3]$. Inconclusive; see §7.

6 The corrected picture

For meromorphic f the period polynomial is canonically a *coset*: the set (6) as (x, t) ranges over the winding lattice. The defect pair $(D_S(r_f), D_U(r_f))$ is an affine-linear image of that lattice; single defect values are class data, the coset is the invariant. The numerical record then organizes itself:

- classes on the axis $D_S = 0$ always exist — e.g. $\tau_0 = i$, now a theorem (Cor. 1(ii));
- classes on the axis $D_U = 0$ existed in every case tested (clean class for the i -block, (10); routed classes for the $1.1i$ -forms; the explicit solution for E_6/j);
- whether the coset contains the *origin* — both relations at once — is the only genuinely open obstruction. For the weight-4 block it does (Proposition 3, a rank-4 lattice of such classes); for the ρ -blocks and the generic pole it is undecided at depth one.

Remark 2 (Isotropy versus singular moduli: what is actually special about i). Two independent features of the elliptic point do different work, and they should not be bundled. *Isotropy*: S fixes i , which (a) forces the parity relation $a_{-1} = ia_{-2}$ between Laurent coefficients, (b) makes $X - iY$ an eigenvector of the slash action ($(X - iY)|S = -i(X - iY)$), collapsing all defect vectors onto single rays, and (c) is what turned Proposition 3 into a two-equation problem with small-integer weights. *Singular moduli*: i is a CM point, so the residue data is arithmetically commensurable (algebraic multiples of powers of π) — this is where the clean rationals like $7/12$ come from. At a generic pole both features die: no parity constraint, generically a simple pole, and defect weights with no rational structure (confirmed at $1.1i$ and at $0.47 + 0.99i$, §5.1). What survives is the exact $\mathbb{Z}[\Gamma]$ -module structure — for a simple pole the entire winding module is

generated by the *single* polynomial $(X - zY)^n$ slashed around the group, $R_{\gamma z} = R_z|\gamma^{-1}$, and its difference/orbit identities hold among transcendental weights on the nose (§5.1). The a-priori expectation was that isotropy is the only mechanism that can create a genuine obstruction to joint exactness (a torsion class in the stabilizer’s cohomology), with generic poles unobstructed at sufficient depth. The weight-6 computations put this in tension: both the ρ -pole and the generic pole stall at index 8 in their windows, while the weight-4 i -block is jointly killable — suggesting the weight, through the Bernoulli arithmetic of the kernel, may govern. Unresolved.

Two inequivalent notions of “canonical class” emerge, and they disagree whenever the residue is nonzero:

1. **The clean class:** geometrically distinguished, essentially unique, defines the L -integral for any meromorphic form; the Eichler–Shimura relations hold up to an explicit anomaly which *is* the residue polynomial (10) — canonical pole data in closed form.
2. **Eichler–Shimura–exact classes:** exact relations, but (where they exist) a full lattice of them, each with a different L^* ; exactness alone does not select a contour, and per-block existence is an arithmetic question.

Which notion the paper should adopt is a judgment call (the natural tie-breakers within (2) being central-value crossing-invariance and the Haberland/Petersson pairing); the data of this note supports stating (1) as the definition and (2) as structure theory of the coset.

7 Open items and caveats

1. **Hand derivation of the E_6/j weight table** $(-11, -3, -8/5, -48/5)$: pending. (The $k = 4$ table is now fully derived, §4.3.) This would make the mod-8 window obstruction of §5 unconditional within its window.
2. **Depth ≥ 2 windings:** run for the generic pole $0.47 + 0.99i$ (§5.1: no joint kill; index-8 relation found); still untested for $E_6/(j - j(1.1i))$ and E_6/j (`t1_depth2.py`, never run). All “no joint solution” statements remain *window-limited*, not proofs.
3. **The index-8 pattern at $k = 6$:** is joint reachability governed by weight (Bernoulli arithmetic of the kernel) rather than pole isotropy? Sharp sub-questions: is 8 provably the lattice index for the depth-two window at $0.47 + 0.99i$ (requires the full relation lattice, not one PSLQ relation); does the index drop with depth; and does $k = 8, 10$ show $8 \rightarrow$ something else.
4. **Speculative** (flagged as such; not verified): since $\mathrm{PSL}_2(\mathbb{Z}) \cong \mathbb{Z}/2 * \mathbb{Z}/3$, the obstruction to an Eichler–Shimura–exact class should be a pair of torsion classes in $H^2(\mathbb{Z}/2; \cdot) \oplus H^2(\mathbb{Z}/3; \cdot)$ with coefficients in the winding module; free orbits (generic poles) would then carry no obstruction, elliptic-orbit poles possibly a finite one. Two identified gaps: the precise identification of the two H^2 components with the two defects, and the module structure of the T -shift vectors Q_p . This predicts a depth- ≥ 2 joint solution for the generic-pole form — a falsifiable test, not yet run.
5. The “66 solutions” count is a float box search cross-checked against the exact ray equations; the exact statement is the Smith-normal-form solution lattice of §4.
6. The $k = 8$ (Δ/E_4) defect geometry (multi-dimensional U -space) is unmapped.

A Numerical methods and scripts

All computations: `mpmath` at 34 significant digits, conventions imported from `wip_meromorphic_periods.py`. Period polynomials of the actual forms by direct `mp.quad` of (1) at $s = 1, \dots, k-1$ (two-segment contour; $\tau_0 = i$ degenerate class or $\tau_0 = 0.3 + 1.4i$ clean class). Residue vectors (5) by circle sums (radius 0.02, 64 points; Hurwitz zeta evaluated directly for the Q_p). Rational identification by PSLQ; integer feasibility by exact Smith normal form on the cleared-denominator ray equations; joint searches by kernel-aware enumeration cross-checked at full precision. Session scripts (scratchpad, in order): `t1_lattice.py` (linear model, first sweeps), `t1_structure.py` (exact ray analysis, PSLQ weights, falsification of [E5], joint-kill box search), `t1_exact.py` (Smith-normal-form verdicts, extended ρ -reps, kernel-aware search), `t1_depth2.py` (depth-2 test, written, *not run*), `t1_generic_047.py` (the §5.1 run: 40 digits, $N_T = 90$ q -terms for the low- Im orbit points, exactly- S -silent move set, box + PSLQ-with-deflation search). Headline residuals: $\mathbf{K}_T|(1+S)$ machine-exact (identity); [E5]-falsifying class 7×10^{-33} ; clean-class U -relation 5.6×10^{-48} ; E_6/j U -kill 4.1×10^{-31} ; PSLQ off-ray residuals $\lesssim 10^{-35}$.

References

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